A prediction should carry its uncertainty

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Bound / ES: límite / PT: limite

An allowed range, not a standard deviation.

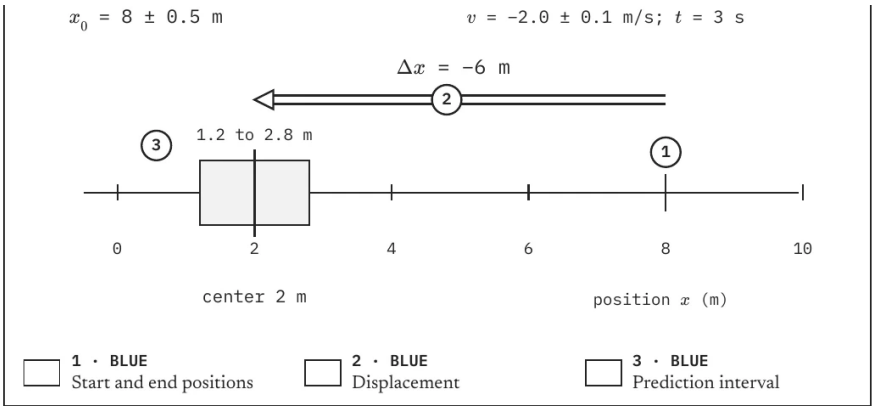
Also on your note page: Initial position, Half-width, Assumption.

B READ THE RELATIONSHIP

Final position = starting position + velocity × time

As you read, find: center · lower end · upper end.

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A cart starts at 8 ± 0.5 m and moves at −2.0 ± 0.1 m/s for an exact 3 s . Find its predicted position interval.

Given x 0 = 8 ± 0.5 m v = −2.0 ± 0.1 m/s t = 3 s ; steady v

Rule Steady v : x f = x 0 + vt . Add input bounds.

Work x f = 8 − 2(3) δ x = .5 + 3(.1) = 0.8 m

Answer 1.2 to 2.8 m 2.0 ± 0.8 m ; full width 1.6 m .

E SAVE FOR THE FINAL MEETING

XJ's position and velocity readings have limits. A forecast should show both the central result and the interval allowed by those limits; the interval does not give an impact probability.

Read the physics: Bound

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Start from the actual starting position. At steady velocity, multiply velocity by time to get the displacement. Add that signed change to the starting position.

02 HOW IT WORKS

Measurements have limits. Mark the central forecast, then the smallest and largest allowed positions. With exact time, allow for the starting-position error and the velocity error multiplied by time.

03 WHAT TO CHECK

Use compatible units before multiplying. A range gives limits; it does not tell you the chance of landing at one particular point.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart starts at 8 ± 0.5 m and moves at -2.0 ± 0.1 m/s for an exact 3 s . Find its predicted position interval. 1.2 to 2.8 m 2.0 ± 0.8 m ; full width 1.6 m .

Model check: Final position = starting position + velocity × time.

05 CONNECT TO XJ

XJ's position and velocity readings have limits. A forecast should show both the central result and the interval allowed by those limits; the interval does not give an impact probability.

SAY IT BEFORE YOU WRITE

What can make the range wider? My center is ___. The allowed positions run from ___ to ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Initial position

ES: posición inicial / PT: posição inicial

Meaning in my words	How can you tell?
How would you explain initial position to a friend? In my words, this means ___.	At 0 s the cart is beside the +5 m mark. Where does its position graph begin? I can tell because ___.
Example or sketch	**Non-example and why**
The cart starts at +5 m and moves 2 m left. What two positions would you mark? My example shows ___.	"The clock starts at zero, so the cart starts at zero." What do the two zeros mean? I would change ___ because ___.

Bound

ES: límite / PT: limite

Meaning in my words	How can you tell?
How would you explain bound to a friend? In my words, this means ___.	For 10 ± 2 km, what is the farthest allowed position on either side? I can tell because ___.
Example or sketch	**Non-example and why**
Mark the bounds for a forecast of 15 ± 3 km. My example shows ___.	A student labels 2 km as the upper endpoint of 10 ± 2 km. What does the 2 mean? I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Half-width

ES: semiancho / PT: meia largura

Meaning in my words	How can you tell?
How would you explain half-width to a friend? In my words, this means ___.	A model allows 8 to 12 km. How far is either endpoint from 10 km? I can tell because ___.
Example or sketch	**Non-example and why**
How would you write a range centered on 20 km with half-width 3 km? My example shows ___.	Someone calls 6 km the half-width of 17 to 23 km. What did they measure? I would change ___ because ___.

Assumption

ES: supuesto / PT: hipótese

Meaning in my words	How can you tell?
How would you explain assumption to a friend? In my words, this means ___.	What must be true to predict another 2 m each second from a cart's earlier motion? I can tell because ___.
Example or sketch	**Non-example and why**
Write an assumption beside a straight-line forecast for XJ. My example shows ___.	Is tomorrow's camera reading an assumption once we measure it? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a center tick and both endpoints. Show how starting-position and velocity bounds contribute to the final half-width.

Labels you can use: center · lower end · upper end.

Choose zero, direction, and scale.



Tell a partner

My center is ___. The allowed positions run from ___ to ___.

How wide should we search?

05

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Read forecast bounds without inventing a probability.

CHOOSE ONE JOB TOGETHER

SURVEY ENGINEER ☐

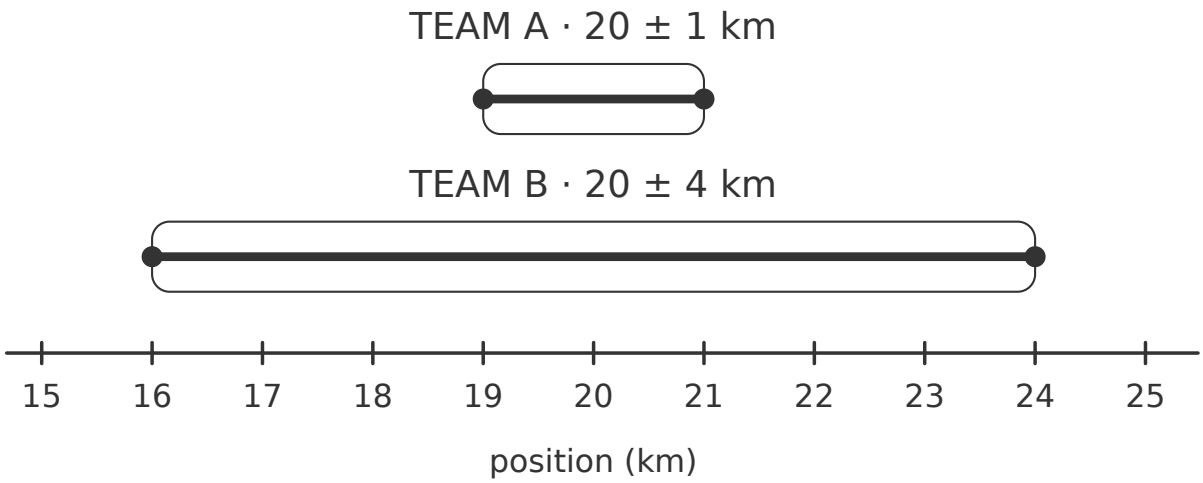
Label both search bands with their endpoints. Recommend which report needs a wider telescope search.

WEATHER-DESK PRESENTER ☐

Present both forecast bands. Explain which search is wider and what the bands do not tell us about impact.

Word help: center / range / endpoints

Two XJ teams predict a center of **20 km**. Team A gives **±1 km**; team B gives **±4 km**. A telescope has one narrow field.



01 OUR FIRST CALL

What search area should the crew prepare for? Use both sets of endpoints.

Team ___ needs a wider search because ___.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the editor's new label. What can you report from these bands?

Now we would say ___ because ___.

Switch speakers. How is changing an input's uncertainty different from changing how you expect XJ to move?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks